Nonreal poles of Weniger's delta transformation for Stieltjes series

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Abstract

The standard (diagonal and near-diagonal) Padé approximants to a Stieltjes series have their poles on the negative real axis, where the function has its branch cut. Borghi (2026) asked whether Weniger's delta transformation, a faster alternative for divergent series, shares this property. We show that in general it does not, and we say exactly when it does. For every Stieltjes measure the denominator has only negative real zeros at orders one and two; at order three this holds exactly when $n + \beta \le (\sqrt{33} - 5)/2$, where n is the starting index and β the shift; from order four on, no choice of n and β protects every measure. The reason is simple: positive moments can be tuned so that only three consecutive coefficients of the denominator matter, and from order four the delta weights fail the classical factor-four test for those three. The nonreal zeros are genuine poles. For one fixed measure we prove that nonreal denominator zeros occur at every sufficiently large order, and we certify genuine nonreal poles exactly at every order from 3 to 30. On the positive side, real-rootedness holds at every order whenever the reciprocal moments form a multiplier sequence, or the moment ratios μ_{r+1}/μ_r increase at least fourfold at each step. Borghi proved real-rootedness for the superfactorial, Laguerre and integer-difference Jacobi families at his normalisation (starting index 0, shift 1); one proof extends these cases to every starting index, shift and order. Exact rational-arithmetic certificates accompany every finite claim.

Keywords: Stieltjes series; Weniger delta transformation; Levin-type transformations; real zeros; multiplier sequences; Padé approximation.
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1 Introduction

Why pole location matters. A Stieltjes function $f(z) = \int_0^\infty (z+t)^{-1}\,\mathrm{d}\mu(t)$ is analytic off the cut $(-\infty, 0]$, and its asymptotic series at infinity often diverges. Padé approximants are the classical tool for summing such series. Their convergence theory rests on one structural fact: the denominators are orthogonal polynomials for μ, so every pole is simple, lies on the cut and carries a positive residue [1, Ch. 5]. Levin-type transformations [11, 19, 15, 5, 7], and in particular Weniger's delta transformation [19], are often much faster than Padé on factorially divergent Stieltjes series [4]; for some hypergeometric data Levin-type approximants even coincide with Padé approximants [16], and Slevinsky derives stable recurrences for the Drummond and factorial Levin-type rational approximants to generalised hypergeometric functions, with results on the placement of their poles [17]. Whether they also place their poles on the cut is therefore the natural first question for any convergence theory of them.

The question. Borghi [2, §3.1] poses it as follows. Writing his equation (23) for the delta transformation, he asks:

> "Is it true that all zeros of the denominator of Equation (23) are confined to the sole negative real axis?"

In §3.3 he specialises: "Henceforth, for the sake of simplicity but without loss of generality, it will be assumed" that the starting index is 0 and the shift is 1. He then writes the denominator Q_k as a $(k-1)$-fold derivative of $z^{k-1}P_k(z)$, where $P_k(z) = \sum_j \binom{k}{j} z^j/\mu_{n+1+j}$ (our (5)), and uses the Gauss–Lucas theorem to argue that differentiation pulls the zeros towards the axis. He states that "only two scenarios are possible: (i) all zeros or (ii) not all zeros of" P_k "are real", that in scenario (i) all zeros of Q_k are real and negative, and that in scenario (ii) "the problem of proving the reality of zeros of" Q_k "remains open". He describes the full claim as "a well-grounded conjecture" and notes that a proof that the convex hull collapses onto the axis "for any Stieltjes sequence" is open [2, §3.3]. His measure class is the one used here: a positive measure on $[0,\infty)$ with infinitely many points of increase and all moments finite. Our β is the parameter γ in the published article's equations (23) and (26); the author preprint uses β in its equations (26) and (29).

The answer. The conjecture is false, already at $n = 0$, $\beta = 1$ and order three. All our counterexamples lie in Borghi's scenario (ii), the case his argument leaves open (Remark 5.3). More precisely, call the pair $(k; n, \beta)$ *universally safe* if $Q_k^{(n,\beta)}$ has only negative real zeros for every admissible measure. Table 1 gives the complete answer.

Table 1: Universal safety of the delta denominator, by order k and shift $b = n + \beta$ (Theorem 4.1). "Unsafe" means some admissible measure gives a genuine nonreal conjugate pair of poles.

Order k	Universally safe	Unsafe
$1, 2$	every $n \geq 0$, $\beta > 0$	never
3	exactly $b \leq b_c = (\sqrt{33} - 5)/2 \approx 0.3723$ (so $n = 0$, $\beta \leq b_c$)	$b > b_c$; e.g. every $n \geq 1$, and $\beta = 1$
≥ 4	never	every $n \geq 0$, $\beta > 0$

The mechanism in one paragraph. The denominator is $Q_k(z) = \sum_j h_j z^j/\mu_{m+j}$ with fixed weights $h_j = \binom{k}{j}(b+j)_{k-1}$ and $m = n + 1$. The explicit measures below make three consecutive normalised moments approach equality in log-convexity while the neighbouring moments diverge, so the moments are huge at every index of the window except three consecutive ones; then Q_k is, up to a negligible error, z^{j-1} times the quadratic $h_{j-1} + h_j z + h_{j+1} z^2$. Such a quadratic has nonreal roots exactly when $h_j^2 < 4h_{j-1}h_{j+1}$. Conversely, if every ratio $h_j^2/(h_{j-1}h_{j+1})$ is at least 4, the classical Hutchinson–Kurtz criterion forces real zeros for every measure. For the delta weights the top ratio is below $10/3$ as soon as $k \geq 4$, and it is below 4 at $k = 3$ exactly when $b > b_c$.

Contributions. Classical inputs are the factor-four criterion [8, 10], Pólya–Schur multiplier sequences [13, 6], accuracy-through-order [7] and Hankel positivity. New here are:

1. the complete order–shift classification (Theorem 4.1), via a realisation lemma: any three consecutive reciprocal-moment coefficients can be isolated by positive measures (Theorem 3.2);

2. proofs that the nonreal zeros are genuine poles; for every fixed n, one measure defeating every shift at order four, and an explicit such measure for $n = 0$ and $n = 1$ (§5);

3. a persistence theorem: $Q_k(w/k^2)$ converges to an entire function E_μ, so nonreal zeros of E_μ force nonreal denominator zeros at all large orders; for one fixed measure this gives nonreal zeros at all large orders, and exact certificates give genuine nonreal poles at every order $3 \le k \le 30$ (§6);

4. a positive theorem (§7). Borghi proved real-rootedness at starting index 0 and shift 1 for the integer superfactorial and Laguerre moments and for Jacobi weights whose parameters differ by an integer [2, §§4.2, 4.3, 4.6]; one operator proof extends these cases to every n, β and k, and to $\Gamma(\nu\ell + c)$ for every $c > 0$. We add practical guidance with an exact diagnostic (§8).

Table 2 lists every result with its exact status.

Table 2: Status of the results. "Proof" means a complete mathematical proof in the text; "certificate" means an exact rational computation in `code/verify_all.py`; "numerical" means floating point, illustrative only.

Result	Status
Classification, Table 1 (Thm 4.1)	proof
Genuine poles in the classification	proof
Explicit examples, $k = 3, 4$; tail family $k = 5$–8 (Table 3)	certificate
One measure, all shifts, $k = 4$, $n \in \{0, 1\}$ (Prop. 5.1)	certificate of a polynomial identity in β
One measure, all shifts, $k = 4$, every fixed n (Cor. 5.2)	proof
Limit $Q_k(w/k^2) \to E_\mu$ (Thm 6.1)	proof
Fixed measure: nonreal zeros at all large k (Thm 6.3)	proof plus one exact certificate
Fixed measure: genuine nonreal poles, $3 \le k \le 30$	certificate
Fixed measure: genuine nonreal poles at every $k \ge 3$ (Conj. 6.4)	open
Positive criteria (Thm 7.2, Cor. 7.3)	proof
Natural families real-rooted on a grid, $\beta \le 5$ (§7)	certificate
All shifts real-rooted iff P_k is (Rem. 7.4)	proof
Pole locations, residues, accuracy (§8)	numerical

Roadmap. Section 2 fixes notation and treats orders one and two. Section 3 proves the reciprocal-moment criterion. Section 4 proves the classification, including genuineness of the poles; the cubic discriminant computation is in Appendix A. Section 5 gives explicit examples. Sections 6–8 contain the persistence theorem, the positive theorem and the practical discussion. Section 9 states the limits of the results and open problems. Appendix B describes the certificates.

2 Setting and low orders

Let μ be a positive Borel measure on $[0, \infty)$ with infinite support and finite moments

$$\mu_\ell = \int_0^\infty t^\ell \, \mathrm{d}\mu(t), \qquad \ell \in \mathbb{N}_0. \tag{1}$$

We call such μ *admissible*. Then $f(z) = \int_0^\infty (z+t)^{-1} \, \mathrm{d}\mu(t) \sim \sum_{\ell \ge 0} (-1)^\ell \mu_\ell z^{-\ell-1}$, with partial sums $f_n(z) = \sum_{\ell=0}^{n} (-1)^\ell \mu_\ell z^{-\ell-1}$. For $k \ge 1$, $n \in \mathbb{N}_0$ and $\beta > 0$ the delta transformation is [19], [2, eq. (23)]

$$\delta_k^{(n)}(\beta; z) = \frac{\Delta^k\{(n+\beta)_{k-1} f_n(z)/\Delta f_n(z)\}}{\Delta^k\{(n+\beta)_{k-1}/\Delta f_n(z)\}}, \tag{2}$$

where Δ acts on n and $(a)_r = a(a+1)\cdots(a+r-1)$. Throughout,

$$b = n + \beta > 0, \qquad m = n + 1.$$

Since $\Delta f_{n+j}(z) = (-1)^{n+j+1}\mu_{n+j+1}z^{-n-j-2}$,

$$\Delta^k\Big\{\frac{(n+\beta)_{k-1}}{\Delta f_n(z)}\Big\} = (-1)^{n+k+1}z^{n+2}Q_k^{(n,\beta)}(z), \qquad Q_k^{(n,\beta)}(z) = \sum_{j=0}^{k}\binom{k}{j}\frac{(b+j)_{k-1}}{\mu_{m+j}}z^j, \tag{3}$$

which is Borghi's eq. (26). The same calculation in the numerator gives

$$\delta_k^{(n)}(\beta; z) = \frac{\sum_{j=0}^{k} a_j z^j f_{n+j}(z)}{Q_k(z)} = \frac{P_{n,k}(z)}{z^{n+1}Q_k(z)}, \qquad a_j = \binom{k}{j}\frac{(b+j)_{k-1}}{\mu_{m+j}}, \tag{4}$$

with $P_{n,k}(z) = \sum_j a_j \sum_{\ell=0}^{n+j}(-1)^\ell \mu_\ell z^{n+j-\ell}$, a real polynomial of degree at most $n+k$. We write

$$P_k(z) = \sum_{j=0}^{k}\binom{k}{j}\frac{z^j}{\mu_{m+j}} \tag{5}$$

for Borghi's auxiliary polynomial [2, eq. (27)]. A zero of Q_k is a *genuine pole* if it is not a zero of $P_{n,k}$, that is, if it is a pole of the reduced rational function.

Cauchy–Schwarz for $t^r\,\mathrm{d}\mu$ gives strict log-convexity,

$$\mu_r\mu_{r+2} > \mu_{r+1}^2 \qquad (r \geq 0), \tag{6}$$

strict because equality would force $t^r\,\mathrm{d}\mu$ to live on one point. All counterexample measures below have moments $\mu_\ell \sim c\,\ell!$, so they satisfy Carleman's condition $\sum \mu_\ell^{-1/(2\ell)} = \infty$ and the moment problem is determinate [14]: the failures are not artefacts of indeterminacy.

Proposition 2.1. *For every admissible measure, Q_1 has one negative zero and Q_2 has two distinct negative zeros.*

Proof. $Q_1 = \mu_m^{-1} + z\mu_{m+1}^{-1}$. For $Q_2 = b\mu_m^{-1} + 2(b+1)\mu_{m+1}^{-1}z + (b+2)\mu_{m+2}^{-1}z^2$, by (6) $\mathrm{Disc}(Q_2) = 4\big((b+1)^2\mu_{m+1}^{-2} - b(b+2)\mu_m^{-1}\mu_{m+2}^{-1}\big) \geq 4\mu_{m+1}^{-2} > 0$. Positive coefficients exclude zeros in $[0,\infty)$. □

3 A reciprocal-moment criterion

Lemma 3.1 (Strict factor-four criterion [8, 10])**.** *If $c_0, \ldots, c_k > 0$ and $c_j^2 > 4c_{j-1}c_{j+1}$ for $1 \leq j \leq k-1$, then $\sum_j c_j z^j$ has k distinct negative zeros.*

Proof. We include the short proof. Put $r_j = c_{j-1}/c_j$, so $r_{j+1} > 4r_j$, and $x_j = \sqrt{r_j r_{j+1}}$, $1 \leq j \leq k-1$. Divide the value at $z = -x_j$ by $(-1)^j c_j x_j^j$. The term of index i becomes $(-1)^{i-j}T_i$ with $T_i = c_i x_j^{i-j}/c_j > 0$ and $T_j = 1$. For $i \geq j$, $T_{i+1}/T_i = x_j/r_{i+1} \leq x_j/r_{j+1} < 1$; for $i \leq j$, $T_{i-1}/T_i = r_i/x_j \leq r_j/x_j < 1$. So both tails are alternating with decreasing moduli, and each is at least $-T_{j\pm1} = -\sqrt{r_j/r_{j+1}} > -1/2$. Hence the polynomial has sign $(-1)^j$ at $-x_j$. As $0 < x_1 < \cdots < x_{k-1}$, the polynomial is positive at 0 and has sign $(-1)^k$ near $-\infty$, these sign changes give k distinct negative zeros. □

Theorem 3.2 (Fixed weights, arbitrary Stieltjes moments)**.** *Fix $m \in \mathbb{N}_0$, $k \geq 1$ and $h_0, \ldots, h_k > 0$. The following are equivalent.*

1. *For every admissible measure, $q_\mu(z) = \sum_{i=0}^{k} h_i z^i/\mu_{m+i}$ has only negative real zeros.*

2. $h_j^2 \geq 4h_{j-1}h_{j+1}$ *for every internal index* $1 \leq j \leq k-1$.

The zeros in 1 are then distinct. If 2 fails, a counterexample can be chosen with a positive exponential density component and moments $\sim c\,\ell!$.

Proof. Sufficiency. For $c_i = h_i/\mu_{m+i}$, $c_j^2/(c_{j-1}c_{j+1}) = \big(h_j^2/(h_{j-1}h_{j+1})\big)\cdot \mu_{m+j-1}\mu_{m+j+1}/\mu_{m+j}^2 > 4$ by (6); apply Lemma 3.1.

Necessity. Suppose $h_j^2 < 4h_{j-1}h_{j+1}$. Put $r_\pm = m + j \pm 1$ and, for $K \geq 2$,

$$\begin{aligned} \mathrm{d}\mu_K^{\mathrm{tri}} &= \mathrm{d}\delta_1 + K^{2r_- - 1}\,\mathrm{d}\delta_{K^{-2}} + K^{-2r_+ - 1}\,\mathrm{d}\delta_{K^2} + K^{-1}e^{-t}\,\mathrm{d}t, \\ \mu_\ell &= 1 + K^{2(r_- - \ell)-1} + K^{2(\ell - r_+)-1} + K^{-1}\ell!. \end{aligned} \tag{7}$$

As $K \to \infty$, $\mu_\ell \to 1$ for $\ell \in \{r_-, r_- + 1, r_+\}$ and $\mu_\ell \to \infty$ for the other $\ell \in [m, m+k]$. Hence $q_\mu \to z^{j-1}(h_{j-1} + h_j z + h_{j+1}z^2)$ coefficientwise, so uniformly on compact sets. The quadratic has two simple nonreal zeros; by Rouché's theorem on small disjoint disks off the real axis, q_μ has nonreal zeros for large K. Each μ_K^{tri} is admissible with $\mu_\ell \sim K^{-1}\ell!$. □

Remark 3.3 (What is classical). Sufficiency is the Hutchinson–Kurtz criterion [8, 10]; on the history and sharpness of the constant 4 see Kostov and Shapiro [9]. Necessity is a realisation statement: three consecutive normalised Stieltjes moments can be made arbitrarily close to the boundary configuration of equal normalised moments while the neighbours dominate. This is in the spirit of classical moment-space geometry [14]; we do not claim it as a new principle, only its use for the delta classification. The preserver theorem of Passare, Rojas and Shapiro [12] concerns a different class (sign-independently real-rooted polynomials): $(1, 8, 16, 8, 1)$ satisfies the inequalities of Theorem 3.2, but $1 + 8z - 16z^2 + 8z^3 + z^4$ has nonreal zeros.

4 Proof of the classification

Theorem 4.1 (Order–shift classification)**.** *Let* $b_{\mathrm{c}} = (\sqrt{33} - 5)/2$ *and* $b = n + \beta$.

1. *If* $b \leq b_{\mathrm{c}}$, *then for every admissible measure* $Q_3^{(n,\beta)}$ *has three distinct negative zeros. The same holds for any positive quadruple* $(\mu_m, \ldots, \mu_{m+3})$ *satisfying the two log-convexity inequalities.*

2. *If* $b > b_{\mathrm{c}}$, *some admissible measure gives* $\delta_3^{(n)}(\beta; \cdot)$ *a genuine nonreal conjugate pair of poles.*

3. *Orders one and two are always safe (Proposition 2.1).*

4. *For every* $k \geq 4$, $n \geq 0$, $\beta > 0$, *some admissible measure gives* $\delta_k^{(n)}(\beta; \cdot)$ *a genuine nonreal conjugate pair of poles.*

In 2 and 4 the measure can be taken with a positive exponential density component, moments $\sim c\,\ell!$, *or (Corollary 4.6) with a smooth positive density.*

The proofs of parts 1–2 and 4 occupy §§4.1–4.3. The measure depends on (k, n, β); §6 addresses one fixed measure across orders.

4.1 Orders four and above

For $h_j = \binom{k}{j}(b+j)_{k-1}$ the top internal ratio is

$$R_k(b) = \frac{h_{k-1}^2}{h_{k-2}h_k} = \frac{2k}{k-1}\,\frac{(b+k-1)(b+2k-3)}{(b+k-2)(b+2k-2)} = \frac{2k}{k-1} + \frac{2k}{(b+k-2)(b+2k-2)}. \tag{8}$$

It decreases in $b > 0$, so for $k \geq 4$

$$R_k(b) < R_k(0) = \frac{k(2k-3)}{(k-1)(k-2)} \leq \frac{10}{3} < 4, \tag{9}$$

the middle inequality being $(k-4)(4k-5) \geq 0$. Theorem 3.2 gives nonreal zeros. For this top triple the right-hand atom of (7) is not needed; the simpler family

$$\mathrm{d}\mu_K^{\mathrm{tail}} = \mathrm{d}\delta_1 + K^{2\ell_0+1}\,\mathrm{d}\delta_{K^{-2}} + K^{-1}e^{-t}\,\mathrm{d}t, \qquad \ell_0 = m+k-3, \tag{10}$$

makes Q_k converge to $z^{k-2}(h_{k-2} + h_{k-1}z + h_k z^2)$. Other internal ratios often fail too (at $k = 8$, $b \to 0$, five of the seven are below 4), so the obstruction is robust.

4.2 The cubic boundary

At $k = 3$ the two internal ratios are $C^2/(BD)$ and $B^2/(CA)$ with $D = b(b+1)$, $C = 3(b+1)(b+2)$, $B = 3(b+2)(b+3)$, $A = (b+3)(b+4)$, and

$$B^2 - 4AC = -3(b+2)(b+3)(b^2+5b-2), \qquad C^2 - 4BD = -3(b+1)(b+2)(b^2+3b-6).$$

The first is nonnegative exactly for $b \leq b_{\mathrm{c}}$, the second for $b \leq b_{\mathrm{c}} + 1$. Hence Theorem 3.2 gives part 1 (with distinct zeros) and the nonreal zeros in part 2. Appendix A gives an independent discriminant proof of part 1 that also covers non-strict log-convex quadruples, and an explicit family realising part 2.

4.3 The nonreal zeros are genuine poles

Lemma 4.2 (Accuracy through order)**.** *For every admissible measure, $\delta_k^{(n)}(\beta; z) - f(z) = O(z^{-n-k-3})$ as $z \to +\infty$. Consequently the Laurent coefficients r_ℓ of the rational function $\delta_k^{(n)}(\beta; z) = \sum_\ell r_\ell z^{-\ell-1}$ at infinity satisfy $r_\ell = (-1)^\ell \mu_\ell$ for $0 \leq \ell \leq n+k+1$.*

Proof. This is the accuracy-through-order property of Levin-type transformations [7, §6.3, Thm. 8]; we check it in the present convention. For fixed j,

$$f_{n+j}(z) - f(z) = (-1)^{n+j}\mu_{n+j+1}z^{-n-j-2} + O(z^{-n-j-3}) \qquad (z \to +\infty),$$

by integrating the finite geometric remainder of $(z+t)^{-1}$. In $\sum_j a_j z^j (f_{n+j} - f)$ the coefficient of z^{-n-2} is $(-1)^n \sum_j (-1)^j \binom{k}{j}(b+j)_{k-1} = 0$, a kth difference of a polynomial of degree $k-1$. Dividing by Q_k (degree k, positive leading coefficient) gives the bound. A rational function has a convergent Laurent expansion at infinity, and an asymptotic expansion along $z \to +\infty$ is unique; so the Laurent coefficients equal the asymptotic ones. □

Proposition 4.3 (Cancellation bound)**.** *For every admissible measure,* $\deg \gcd(P_{n,k}, Q_k^{(n,\beta)}) < \lceil k/2 \rceil$. *In particular at orders three and four no nonreal conjugate pair can cancel.*

Proof. Suppose cancellation leaves a monic denominator factor $S(z) = \sum_{i=0}^r s_i z^i$ with $r \leq k/2$, so that $\delta = U/(z^m S)$ with $\deg U \leq m + r - 1$. Since $S\delta = U/z^m$ has no terms below z^{-m}, $\sum_i s_i r_{m+i+j} = 0$ for $0 \leq j \leq r$. As $m + 2r \leq m + k$, Lemma 4.2 turns this into $(\mu_{m+i+j})_{i,j=0}^r (s_0, -s_1, \ldots, (-1)^r s_r)^T = 0$. This Hankel matrix is positive definite, its quadratic form being $\int t^m (\sum_i v_i t^i)^2 \,\mathrm{d}\mu > 0$ for $v \neq 0$; contradiction with $s_r = 1$. A nonreal zero can only cancel together with its conjugate. □

For $k \geq 5$ we exclude cancellation by moving one moment.

Lemma 4.4 (Finitely many bad last moments)**.** *Fix $\mu_0, \ldots, \mu_{n+k} > 0$. As μ_{n+k+1} varies over $(0, \infty)$, $P_{n,k}$ and Q_k have a common zero for at most $k-1$ values.*

Proof. Put $N = n+k$ and $Q_k = A + a_k z^k$, where only a_k depends on μ_{N+1}, injectively. At a common zero $r \neq 0$, eliminating $a_k r^k = -A(r)$ gives $H(r) = 0$ for $H(z) = z^{N+1} \sum_{j<k} a_j z^j (f_{n+j}(z) - f_N(z))$, a polynomial with $\deg H \leq k-1$ and $H(0) = -a_0(-1)^N \mu_N \neq 0$. Each of its at most $k-1$ zeros determines $a_k = -A(r)/r^k$. □

Lemma 4.5 (Moving one moment inside the admissible class)**.** *Let μ contain $\varepsilon e^{-t}\,\mathrm{d}t$, $\varepsilon > 0$, and fix $L \geq 1$. With the shifted Legendre polynomial $\psi_L(t) = \frac{1}{L!}\frac{\mathrm{d}^L}{\mathrm{d}t^L}[(t-1)^L(t-2)^L]$, the measures $\mu + \theta\psi_L \mathbf{1}_{[1,2]}\,\mathrm{d}t$, $|\theta|\,\|\psi_L\|_\infty < \varepsilon e^{-2}$, are admissible, have the same moments of index $< L$, and have Lth moment $\mu_L + \theta(L!)^2/(2L+1)!$. Factorial asymptotics are preserved.*

Proof. Integration by parts L times gives $\int_1^2 t^j \psi_L = 0$ for $j < L$ and $\int_1^2 t^L \psi_L = (L!)^2/(2L+1)!$. The density stays positive, and the change of the ℓth moment is $O(2^\ell)$. □

Proof of Theorem 4.1, parts 2 and 4. Part 2: the construction of Appendix A and Proposition 4.3. Part 4: the family (10) gives a simple nonreal pair for large K; it persists under small coefficient changes (Rouché). Apply Lemma 4.5 with $L = n+k+1$ and choose the last moment close to its old value and outside the finite set of Lemma 4.4. Then $P_{n,k}$ and Q_k are coprime. □

Corollary 4.6 (Smooth densities)**.** *Every existence statement of Theorem 4.1 holds with measures having a C^∞ strictly positive density on $(0,\infty)$ and moments $\sim c\,\ell!$.*

Proof. Take a coprime counterexample as above and convolve it with a C^∞ probability density φ on $\mathbb{R}$ supported in $[0,\eta]$ and positive on $(0,\eta)$. The result has a C^∞ density, positive on $(0,\infty)$ (the exponential part already is); the indicator-cut Legendre term becomes smooth after convolution. Finitely many moments move continuously as $\eta \downarrow 0$, and nonreal zeros in fixed disks and a nonzero resultant are open conditions. Beyond a compact set the density is a positive multiple of e^{-t}. □

5 Explicit examples

All integers and identities in this section are recomputed by `code/verify_all.py` and listed in `results/verification.json`; we display only what a reader needs.

Order three, $n = 0$, $\beta = 1$. For $\mathrm{d}\mu = \mathrm{d}\delta_1 + 50\,\mathrm{d}\delta_{1/50} + 10^{-3}e^{-t}\,\mathrm{d}t$, $\mu_\ell = 1 + 50^{1-\ell} + \ell!/1000$, the denominator is proportional to

$$\widehat{Q}(z) = 1071932365000z^3 + 1963236457665z^2 + 966634623774z + 54855937892,$$

whose discriminant is a negative integer of 47 digits. Exact Euclidean division gives $\gcd(\widehat{P}, \widehat{Q}) = 1$, with $\delta_3^{(0)}(1; z) = \widehat{P}(z)/(1000z\widehat{Q}(z))$. The poles are shown in Figure 1.

Order four inside the cubic-safe range. For $\mathrm{d}\lambda = \mathrm{d}\delta_1 + 10^5\,\mathrm{d}\delta_{1/100} + 10^{-3}e^{-t}\,\mathrm{d}t$ at $n = 0$, $\beta = 1/3 < b_\mathrm{c}$, every cubic is real-rooted, but the quartic denominator $\widehat{Q}_4$ violates the Laguerre inequality $(q')^2 - qq'' \geq 0$ (valid at every real point for real-rooted q, since $(q')^2 - qq'' = q^2 \sum_r (x-r)^{-2}$): the value at $x = -1$ is a negative 38-digit integer. Numerically, the two real zeros lie in $(-0.03, 0)$ and the nonreal pair is near $-1.14 \pm 0.23i$, so $x = -1$ is close to the pair. Exact arithmetic gives $\delta_4^{(0)}(1/3; z) = 3\widehat{P}_4(z)/(1000z\widehat{Q}_4(z))$ with primitive integer polynomials $\widehat{P}_4, \widehat{Q}_4$ (listed in the certificate) and $\gcd(\widehat{P}_4, \widehat{Q}_4) = 1$.

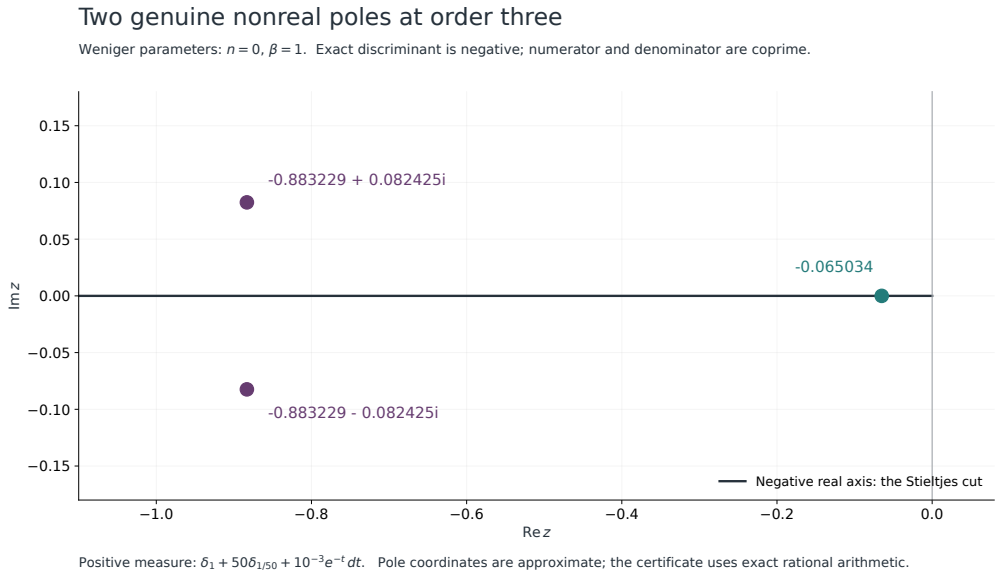


Figure 1: Poles of $\delta_3^{(0)}(1; z)$ for $\mathrm{d}\mu = \mathrm{d}\delta_1 + 50\,\mathrm{d}\delta_{1/50} + 10^{-3}e^{-t}\,\mathrm{d}t$: one negative real pole, the pole at the branch point $z = 0$ (not marked; residue about 32.7, §8), and a genuine nonreal pair. Coordinates are floating-point approximations; the nonreal pair is certified by the exact discriminant.

One input, every shift.

Proposition 5.1. *For the measure λ and every $\beta > 0$, $Q_4^{(n,\beta)}$ has exactly one nonreal conjugate pair of zeros, and it is a genuine pair of poles of $\delta_4^{(n)}(\beta; \cdot)$, both for $n = 0$ and for $n = 1$.*

Proof. For $n = 0$ the discriminant of $Q_4^{(0,\beta)}$, computed over $\mathbb{Q}[\beta]$ by the 16-term quartic formula and independently by the 7×7 Sylvester determinant, is a polynomial of degree 18 in β with all nineteen coefficients negative; it factors as $-c(\beta+1)(\beta+2)^2(\beta+4)^2(\beta+5)\,C(\beta)$ with $c > 0$ and C of degree 12 with positive integer coefficients (listed in the certificate). For $n = 1$ the discriminant, as a polynomial in β of degree 18, again has only negative coefficients. A negative quartic discriminant means exactly one nonreal pair; positive coefficients make the real zeros negative; Proposition 4.3 excludes cancellation. □

The same measure fails at $n = 2$: there the discriminant is positive at $\beta = 0$, and at $\beta = 1$ an exact Sturm count shows that $Q_4^{(2,1)}$ has four real zeros, all negative, so $\delta_4^{(2)}(1; \cdot)$ has no nonreal poles. The present explicit measure does not work for every n; the following construction permits the measure to depend on n. For each fixed n it gives such a measure, though without explicit constants.

Corollary 5.2 (One quartic measure for all shifts)**.** *For every fixed $n \geq 0$ there is an admissible measure, with moments $\sim c\,\ell!$, such that $\delta_4^{(n)}(\beta; \cdot)$ has a genuine nonreal conjugate pair of poles for every $\beta > 0$.*

Proof. Take μ_K^{tail} of (10) with $k = 4$, so $\ell_0 = m + 1$ and $\mu_\ell = 1 + K^{2m+3-2\ell} + K^{-1}\ell!$; it does not depend on β. Write $b = n + \beta \in (n, \infty) \subset [0, \infty)$ and divide $Q_4^{(n,\beta)}$ by its leading coefficient h_4/μ_{m+4}. The resulting coefficients are $(h_j/h_4)(\mu_{m+4}/\mu_{m+j})$. Here $0 < h_j/h_4 \leq \binom{4}{j}$, since $(b+j)_3 \leq (b+4)_3$. The moment ratios do not depend on b: they tend to 0 for $j = 0, 1$ (as $\mu_m, \mu_{m+1} \geq K$) and to 1 for $j = 2, 3, 4$. Hence the normalised denominator converges to $z^2 g_b(z)$,

coefficientwise and uniformly in $b \geq 0$, where

$$g_b(z) = z^2 + \frac{4(b+3)}{b+6}z + \frac{6(b+2)(b+3)}{(b+5)(b+6)}, \qquad \operatorname{Disc}(g_b) = -\frac{8(b+3)(b^2+8b+6)}{(b+5)(b+6)^2} \leq -\frac{4}{5};$$

the inequality is $10(b+3)(b^2+8b+6) - (b+5)(b+6)^2 = 9b^3 + 93b^2 + 204b \geq 0$. The zeros of g_b are $w_b, \bar{w}_b$ with $\operatorname{Re} w_b = -2(b+3)/(b+6) \in [-2,-1]$ and $\operatorname{Im} w_b = \frac{1}{2}\sqrt{-\operatorname{Disc}(g_b)} \geq 1/\sqrt{5}$. On the circle $|z - w_b| = \rho := 1/(2\sqrt{5})$ we have $|z| \geq 1 - \rho$ and $|z - \bar{w}_b| \geq 2/\sqrt{5} - \rho$, so $|z^2 g_b(z)| \geq (1-\rho)^2\rho(2/\sqrt{5} - \rho) =: \varepsilon > 0$, uniformly in b. All these circles lie in $|z| \leq 3$, where the difference between the normalised denominator and $z^2 g_b$ is at most $3^4 \cdot 5$ times the largest coefficient error, which tends to 0 uniformly in b. Fix K so large that this bound is below ε. By Rouché's theorem, for every $b \geq 0$ the denominator has a zero in the disk $|z - w_b| < \rho$, which lies in the upper half-plane, together with its conjugate. Proposition 4.3 excludes cancellation of a nonreal pair at order four, for every β. The moments satisfy $\mu_\ell \sim K^{-1}\ell!$. □

Explicit sizes for orders five to eight. Table 3 replaces the limit $K \to \infty$ in (10) by explicit values: for each case the smallest $K = 2^e$ ($e \leq 16$) is listed at which an exact Sturm count shows nonreal zeros of Q_k and exact Euclidean division shows $\gcd(P_{n,k}, Q_k) = 1$. No perturbation is needed.

Table 3: Certified genuine nonreal poles from μ_K^{tail}, eq. (10): smallest $K = 2^e$, and the number of nonreal zeros of Q_k.

	$n=0$			$n=1$		
k	$\beta=1/10$	$\beta=1$	$\beta=5$	$\beta=1/10$	$\beta=1$	$\beta=5$
5	2^{11} (2)	2^{10} (2)	2 (2)	4 (2)	4 (2)	2 (2)
6	4 (2)	4 (2)	2 (4)	2 (2)	2 (2)	2 (2)
7	2 (2)	2 (2)	2 (4)	2 (2)	2 (2)	2 (4)
8	2 (2)	2 (4)	2 (4)	2 (2)	2 (4)	2 (6)

Remark 5.3 (Borghi's scenario (ii)). In every counterexample of this paper Borghi's polynomial P_k has nonreal zeros (exact Sturm counts: 2 for the cubic and quartic examples, 2–6 in Table 3). This must be so: by Theorem 7.2(a) below, real-rooted P_k forces real-rooted Q_k. The counterexamples therefore lie exactly in the case his Gauss–Lucas argument leaves open, and show that the open case can fail. It need not fail. Borghi already shows this numerically: his Fig. 1 [2, §4.6] displays, for a Jacobi weight whose parameters do not differ by an integer ($k = 20$), a P_k with nonreal zeros and a Q_k with only real zeros. The non-integer Jacobi tests in §7 show the same behaviour at other (k, n, β).

6 One fixed measure at all large orders

Theorem 4.1 lets the measure depend on the order. Here we fix the measure.

Theorem 6.1 (Small-scale limit). *Let μ be admissible, $n \geq 0$, $\beta > 0$. Define*

$$E_\mu(w) = \sum_{j \geq 0} \frac{w^j}{j!\,(b)_j\,\mu_{m+j}}. \tag{11}$$

Then E_μ is entire of order at most $1/2$, and $(b)_{k-1}^{-1} Q_k^{(n,\beta)}(w/k^2) \to E_\mu(w)$ uniformly on compact subsets of $\mathbb{C}$ as $k \to \infty$.

Proof. Using $(b+j)_{k-1}/(b)_{k-1} = (b+k-1)_j/(b)_j$,

$$\frac{Q_k(w/k^2)}{(b)_{k-1}} = \sum_{j=0}^{k} t_{k,j}\frac{w^j}{j!(b)_j\mu_{m+j}}, \qquad t_{k,j} = \prod_{i=0}^{j-1}\Big(1-\frac{i}{k}\Big)\Big(1+\frac{b-1+i}{k}\Big).$$

For $1 \le i < k$ the factor equals $1+(b-1)/k-i(b-1+i)/k^2 \in [0, 1+|b-1|]$ (note $b-1+i \ge b > 0$), and it equals $1+(b-1)/k$ for $i=0$; so $0 \le t_{k,j} \le (1+|b-1|)^j$ and $t_{k,j} \to 1$ for fixed j. By (6), $\mu_{m+j} \ge \mu_m(\mu_{m+1}/\mu_m)^j$, so the coefficients of E_μ are at most $C\theta^j/(j!(b)_j)$; this gives order at most $1/2$ and a summable majorant on every disk, and the Weierstrass M-test gives locally uniform convergence. □

Corollary 6.2. *Let μ, n, β be fixed.*

1. *If E_μ has a nonreal zero, there is k_0 such that $Q_k^{(n,\beta)}$ has a nonreal zero for every $k \ge k_0$; these zeros are asymptotic to w^*/k^2 for a nonreal zero w^* of E_μ.*
2. *If $Q_k^{(n,\beta)}$ is real-rooted for infinitely many k, then E_μ belongs to the Laguerre–Pólya class.*
3. *E_μ has a nonreal zero as soon as one Jensen polynomial $g_d(w) = \sum_{j=0}^{d}\binom{d}{j}w^j/((b)_j\mu_{m+j})$ has a nonreal zero.*

Proof. 1 and 2 follow from Theorem 6.1 and Hurwitz's theorem (the Laguerre–Pólya class is closed under locally uniform limits of real-rooted polynomials). For 3: a real entire function of order < 2 with only real zeros is, by Hadamard's theorem, in the Laguerre–Pólya class, and then all its Jensen polynomials are real-rooted [13], [6, §1]. □

Theorem 6.3 (Persistence for one measure)**.** *Let* $\mathrm{d}\mu = \mathrm{d}\delta_1 + 50\,\mathrm{d}\delta_{1/50} + 10^{-3}e^{-t}\,\mathrm{d}t$, $n=0$, $\beta=1$.

1. *For every $3 \le k \le 30$, $Q_k^{(0,1)}$ has nonreal zeros and $\gcd(P_{0,k}, Q_k) = 1$; so $\delta_k^{(0)}(1;\cdot)$ has genuine nonreal poles. The numbers of nonreal zeros are given in Table 4.*
2. *There is k_0 such that $Q_k^{(0,1)}$ has nonreal zeros for every $k \ge k_0$. In particular nonreal denominator zeros occur at infinitely many orders.*

Proof. Part 1 is an exact Sturm-sequence count and an exact gcd at each order (certificate, section `D`). For part 2, here $E_\mu(w) = \sum_j w^j/((j!)^2\mu_{j+1})$, and the Jensen polynomial $g_6(w) = \sum_{j=0}^{6}\binom{6}{j}w^j/(j!\,\mu_{j+1})$ has exactly two nonreal zeros by an exact Sturm count ($g_1, \dots, g_5$ are real-rooted). Apply Corollary 6.2. □

Table 4: Exact numbers of nonreal zeros of $Q_k^{(0,1)}$ (row Q) and of Borghi's P_k (row P) for the measure of Theorem 6.3; in every case $\gcd(P_{0,k}, Q_k) = 1$.

k	3	4–7	8	9	10	11–12	13–14	15–17	18–19	20	21–23	24–29	30
Q	2	2	4	4	4	6	6	8	8	10	10	10	10
P	2	2	4	6	8	8	10	12	14	14	16	18	20

Numerically, E_μ has the zero $w^* \approx -9.1021 + 9.1273\,i$, and the zero of Q_k closest to w^*/k^2 satisfies $k^2 z \approx -10.12 + 8.04i$, $-9.38 + 8.88i$, $-9.23 + 9.02i$ at $k = 10, 20, 30$. This nonreal pair of denominator zeros thus approaches the branch point $z=0$ at rate k^{-2} (it is certified to give genuine poles only for $k \le 30$). The proof of part 2 is not effective: it does not give k_0, so we do not close the gap. Nor does Proposition 4.3 exclude cancellation at large k.

Conjecture 6.4. *For the measure of Theorem 6.3, $\delta_k^{(0)}(1;\cdot)$ has genuine nonreal poles at every order $k \ge 3$.*

The evidence is Table 4, part 2 of the theorem, and the convergence of the scaled zeros to w^*.

7 A positive theorem and natural families

Lemma 7.1 (Rolle step)**.** *Let f be a real polynomial of degree d with positive coefficients and only real zeros, and $c > 0$. Then $zf'(z) + cf(z)$ has degree d and only negative real zeros.*

Proof. For $x < 0$ put $g(x) = (-x)^c f(x)$; then $g'(x) = -(-x)^{c-1}(xf'(x) + cf(x))$. Let f have distinct zeros $s_1 < \cdots < s_p < 0$ with multiplicities summing to d. A zero of multiplicity r of f is a zero of multiplicity $r-1$ of $xf' + cf$. Rolle's theorem gives a zero of g' in each (s_i, s_{i+1}) and one in $(s_p, 0)$, because $g(s_p) = 0 = g(0^-)$. This accounts for d zeros, all negative. □

Recall that a real sequence $(\gamma_j)_{j\ge 0}$ is a *multiplier sequence* if $\sum \gamma_j a_j z^j$ is real-rooted whenever $\sum a_j z^j$ is. By Pólya and Schur [13] (see [6]), a positive sequence is a multiplier sequence if and only if $\sum_{j\le k} \binom{k}{j}\gamma_j z^j$ is real-rooted for every k.

Theorem 7.2 (Positive criteria)**.** *Fix $n \ge 0$, $m = n+1$, $k \ge 1$ and an admissible measure.*

(a) If Borghi's P_k in (5) has only real zeros, then $Q_k^{(n,\beta)}$ has only negative real zeros for every $\beta > 0$.

(b) If $(1/\mu_{m+j})_{j\ge 0}$ is a multiplier sequence (equivalently, P_k is real-rooted for every k), then $Q_k^{(n,\beta)}$ has only negative real zeros for every $k \ge 1$ and $\beta > 0$.

(c) If $\mu_{r-1}\mu_{r+1} \ge 4\mu_r^2$ for $m+1 \le r \le m+k-1$, then $Q_k^{(n,\beta)}$ has k distinct negative zeros for every $\beta > 0$.

Proof. (a) With $\vartheta = z\,\mathrm{d}/\mathrm{d}z$ we have $(\vartheta + c)z^j = (j+c)z^j$, so $Q_k = \prod_{i=0}^{k-2}(\vartheta + b + i)\,P_k$; apply Lemma 7.1 $k-1$ times. This is Borghi's identity $Q_k = z^{1-b}\frac{\mathrm{d}^{k-1}}{\mathrm{d}z^{k-1}}[z^{b+k-2}P_k]$ in operator form, and it extends his scenario (i) to every n and β (non-integer exponents need Lemma 7.1). (b) follows from (a) and the Pólya–Schur characterisation. (c) For the delta weights,

$$\frac{h_j^2}{h_{j-1}h_{j+1}} = \frac{(j+1)(k-j+1)}{j(k-j)}\Big(1 + \frac{k-1}{(s-1)(s+k-1)}\Big) > 1, \qquad s = b + j > 1,$$

so the coefficient ratios of Q_k exceed 4; apply Lemma 3.1. □

Corollary 7.3 (Natural families)**.** *$Q_k^{(n,\beta)}$ has only negative real zeros for every k, n and $\beta > 0$ in each of the following cases.*

1. *$\mu_\ell = \Gamma(\ell + a)$, $a > 0$: the Euler series ($a = 1$), $\Gamma(\ell + 1/2)$, $\Gamma(\ell + 5)$, and Borghi's Laguerre distribution [2, §4.3]. Here $P_k(z) = \frac{k!}{\Gamma(m+a+k)}L_k^{(m+a-1)}(-z)$, and Laguerre polynomials with parameter > -1 have positive simple zeros [18, Thm. 3.3.1]. With $a = 1$, $n = 0$, $\beta = 1$ this recovers the real-rootedness of the Euler-series denominator quoted by Borghi from [4, Thm. 5.1].*
2. *$\mu_\ell = \Gamma(\nu\ell + c)$ with integer $\nu \ge 1$ and $c > 0$, for example $(2\ell)!$ and $(3\ell)!$.*
3. *Beta (Jacobi-type) weights $t^\alpha(1-t)^p$ on $[0,1]$ with $\alpha > -1$ and integer $p \ge 0$, including $\mu_\ell = 1/(\ell+1)$.*
4. *Log-normal moments $\mu_\ell = e^{\sigma^2\ell^2/2}$ with $\sigma^2 \ge 2\log 2$ (distinct zeros).*

Proof. 1: by (a) of Theorem 7.2 for every k; in particular $(1/\Gamma(j+a'))_j$ is a multiplier sequence for every $a' > 0$. 2: Gauss's multiplication formula gives $1/\mu_{m+j} = C\nu^{-\nu j}\prod_{i=0}^{\nu-1} 1/\Gamma(j+m+(c+i)/\nu)$, a termwise product of multiplier sequences, hence a multiplier sequence; apply (b). 3: $1/\mu_{m+j}$ is a positive constant times $(\alpha+m+1+j)_{p+1}$, a polynomial in j with negative zeros, so $P_k = \prod_{i=0}^{p}(\vartheta + \alpha + m + 1 + i)(1+z)^k$ up to a constant, which is real-rooted by Lemma 7.1; apply (a). 4: the moment ratio equals $e^{\sigma^2} \ge 4$; apply (c). □

At Borghi's normalisation $n = 0$, $\beta = 1$, the cases of items 1–3 are due to him, except item 2 with $c \geq 2$: he proves real-rootedness of P_k, and hence of Q_k, for the superfactorial moments $\Gamma(\nu\ell + q + 1)$ with integer ν and $q \in (-1, 1)$ through Pólya frequency functions [2, §4.2], for the Laguerre distribution as a special case [2, §4.3], and for Jacobi weights whose parameters differ by an integer [2, §4.6] (our item 3). What is new in Corollary 7.3 is the extension to every n, β and k (and to every $c > 0$ in item 2), obtained by one proof through Theorem 7.2. The limit function of §6 is consistent with this picture: when $(1/\mu_{m+j})$ is a multiplier sequence, E_μ is in the Laguerre–Pólya class.

Exact tests. The certificate (section `F`) counts real zeros exactly for $4 \leq k \leq 8$, $n \in \{0, 1, 2\}$ and $\beta \in \{1/10, 1/2, 1, 2, 5\}$ (975 Sturm counts) for: $\ell!$, $(2\ell)!$, $(3\ell)!$, $\Gamma(\ell + 1/2)$, $\Gamma(\ell + 5)$, $1/(\ell+1)$, log-normal 2^{ℓ^2}, the Weniger–Čížek moments $\mu_\ell = (\frac{1}{2} + \nu)_\ell(\frac{1}{2} - \nu)_\ell/(2^\ell \ell!)$ of K_ν [20] for $\nu \in \{0, 1/4, 1/3\}$ (Borghi's eq. (47) case, for which he reports no proof), and Jacobi weights with $(\alpha, p) \in \{(-\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, -\frac{1}{2}), (\frac{1}{3}, \frac{2}{3})\}$. Every Q_k on this grid is real-rooted. For the K_ν and Jacobi cases this is evidence, not proof, and only for the tested shifts $\beta \leq 5$; for the three non-integer Jacobi weights Borghi's P_k has nonreal zeros at every tested (k, n), so these are real-rooted cases inside scenario (ii), as in Borghi's Fig. 1. By Remark 7.4 they cannot stay real-rooted for every shift.

Remark 7.4 (Large shifts). Fix n, k and an admissible measure. Then $\beta^{-(k-1)} Q_k^{(n,\beta)}(z) \to P_k(z)$ as $\beta \to \infty$, so $Q_k^{(n,\beta)}$ is real-rooted for every $\beta > 0$ if and only if P_k is real-rooted.

Proof. With $b = n + \beta$, the coefficient of z^j in $\beta^{-(k-1)} Q_k$ is $\binom{k}{j}((b + j)_{k-1}/\beta^{k-1})/\mu_{m+j}$, and $(b + j)_{k-1}/\beta^{k-1} = \prod_{i=0}^{k-2}(1 + (n + j + i)/\beta) \to 1$. So the coefficients converge to those of P_k, and the degree stays k with leading coefficient bounded away from 0. If P_k has a nonreal zero z^*, Hurwitz's theorem on a small disk about z^* in the open half-plane gives a nonreal zero of $Q_k^{(n,\beta)}$ for all large β. Conversely, the zeros of $Q_k^{(n,\beta)}$ converge to those of P_k with multiplicity, so P_k is real-rooted if $Q_k^{(n,\beta)}$ is real-rooted along a sequence $\beta \to \infty$. The reverse implication is Theorem 7.2(a). □

For the Jacobi weight $t^{-1/2}(1 - t)^{1/2}$, $n = 0$, $k = 4$ and $\beta = 20$, outside the tested grid, the primitive denominator is $416000z^4 + 1104000z^3 + 1020096z^2 + 371910z + 40425$; its discriminant is negative, an exact Sturm count gives two nonreal zeros, and $\gcd(P_{0,4}, Q_4) = 1$, so these are genuine nonreal poles (certificate, section `F`).

8 Practical consequences

The figures in this section are floating-point computations (mpmath, 60 digits) and are illustrative; only the counts of nonreal zeros and the gcd degrees are exact.

Where the poles sit and what they do. In the cubic example (Figure 1) the nonreal pair $-0.8832 \pm 0.0824i$ lies close to the cut and has residues $0.357 \pm 1.360i$, of modulus about 1.4; the real poles at -0.0650 and 0 have positive residues 17.6 and 32.7. These are not small Froissart-type residues, and they are not positive, so δ_3 is not itself a Stieltjes function, unlike every Padé approximant [1, Ch. 5]. The relative error of δ_3 is 1.1×10^{-8} at $z = 10$, 1.1×10^{-4} at $z = 1$, 2.9×10^{-4} at $z = i$, 3.9×10^{-2} at $z = 0.1$ and 1.3×10^{-2} at $z = -0.5 + 0.3i$: the defect is local to the region near the cut. In the persistence example the nonreal denominator zeros move to the branch point like k^{-2}; they are certified genuine poles for $3 \leq k \leq 30$ only.

Accuracy co-occurrence. For $f(z) = 100/(z+2) + \int_0^\infty e^{-t}(z+t)^{-1}\,\mathrm{d}t$ (moments $100 \cdot 2^\ell + \ell!$; a continuum plus one pole), $n = 0$, $\beta = 1$:

k	2	4	8	12	16	20	30	40	50
relative error at $z = 1$	1.8e−3	1.5e−2	0.43	1.55	1.52	0.98	0.43	4.5e−2	9.5e−3
nonreal zeros of Q_k (exact)	0	0	2	4	8	10	14	16	18

with $\gcd(P_{0,k}, Q_k) = 1$ in every case, while the pure Euler series has errors 2.2×10^{-7} ($k = 8$) and 6.5×10^{-18} ($k = 30$). The error transient and the nonreal poles occur together; we make no causal claim, since the poles stay at distance of order one from $z = 1$.

A diagnostic. For given data, real-rootedness of the computed Q_k can be decided exactly: clear denominators and count real zeros with a Sturm sequence (or the sign of the discriminant at $k \le 4$); decide genuineness with an exact gcd. The function `real_root_count` in `code/verify_all.py` does this in rational arithmetic. The factor-four test of Lemma 3.1 is *not* a useful diagnostic: for the Euler moments at $n = 0$, $\beta = 1$, $k = 10$ its smallest ratio is $1.82 < 4$, yet Q_{10} is real-rooted. It is a sufficient condition only.

9 Scope and open problems

The classification concerns pole location at fixed finite order for the worst case over all admissible measures. It says nothing against the convergence results for special families [4, 3], and Section 7 shows that those families are real-rooted at every order. These results rule out an argument requiring the unreduced delta denominator to be real-rooted at all sufficiently large orders. Persistence of genuine nonreal poles for the fixed measure remains open beyond the certified orders ($3 \le k \le 30$; Conjecture 6.4). The practical lesson is that, for Stieltjes series of mixed character (for example a continuum plus isolated poles), real-rootedness of the delta denominator should be checked rather than assumed.

Open problems: (i) prove Conjecture 6.4, including genuineness at large orders; (ii) characterise the measures for which E_μ lies in the Laguerre–Pólya class, and decide whether this is also sufficient for real-rootedness at all large orders; (iii) decide the Weniger–Čížek K_ν family; (iv) treat other Levin-type weights, such as Levin's t and u transformations.

A The cubic boundary by discriminants

With $m = n+1$, set $\gamma = \mu_{m+1}/\mu_m$ and normalised moments $\rho_r = \mu_{m+r}/(\mu_m \gamma^r)$, so $(\rho_0, \dots, \rho_3) = (1, 1, u, u^2 v)$ with $u = \mu_m \mu_{m+2}/\mu_{m+1}^2 \ge 1$, $v = \mu_{m+1}\mu_{m+3}/\mu_{m+2}^2 \ge 1$. With $x = 1/u$, $y = 1/v$ and D, C, B, A as in §4.2,

$$p(\zeta) := \mu_m Q_3(\gamma\zeta) = D + C\zeta + Bx\zeta^2 + Ax^2 y\zeta^3, \qquad 0 < x, y \le 1.$$

Using $\mathrm{Disc}(e_3\zeta^3 + e_2\zeta^2 + e_1\zeta + e_0) = e_2^2 e_1^2 - 4e_3 e_1^3 - 4e_2^3 e_0 - 27e_3^2 e_0^2 + 18 e_3 e_2 e_1 e_0$,

$$\mathrm{Disc}(p) = x^2 F(x, y), \qquad F = B^2C^2 - 4AC^3 y + x(-4B^3 D + 18ABCDy) - 27A^2 D^2 x^2 y^2. \quad (12)$$

(The certificate verifies (12) as an identity in $\mathbb{Q}[b, x, y]$.) F is concave in x for fixed y and in y for fixed x, so $F(x, y) \ge (1-x)(1-y)F(0,0) + (1-x)yF(0,1) + x(1-y)F(1,0) + xyF(1,1)$, with $F(0,0) = B^2C^2$, $F(0,1) = C^2(B^2 - 4AC)$, $F(1,0) = B^2(C^2 - 4BD)$ and $F(1,1) = 432(b+1)(b+3)$ (the last from $D + C\zeta + B\zeta^2 + A\zeta^3 = (1+\zeta)[A\zeta^2 + 2(b+1)(b+3)\zeta + D]$). For $b \le b_c$ all four are nonnegative and the last positive, so $F \ge xyF(1,1) > 0$ and p has three distinct negative zeros; this includes $b = b_c$ and non-strict log-convex quadruples.

For $b > b_c$, $F(0,1) < 0$. The admissible family $\mathrm{d}\mu_K^{\mathrm{cub}} = \mathrm{d}\delta_1 + K^{2m+1}\,\mathrm{d}\delta_{K^{-2}} + K^{-1}e^{-t}\,\mathrm{d}t$, $\mu_\ell = 1 + K^{2m+1-2\ell} + K^{-1}\ell!$, has $\mu_m \sim K$ and $\mu_{m+1}, \mu_{m+2}, \mu_{m+3} \to 1$, so $(x, y) \to (0, 1)$ and $F < 0$ for large K. The certificate lists the first $K = 2^e$ that works for four parameter pairs (for example $K = 128$ at $n = 0$, $\beta = 1$, and $K = 4096$ at $n = 0$, $\beta = 2/5$, close to the threshold).

B Reproducibility

The package is archived at https://doi.org/10.5281/zenodo.23586255; the repository is https://github.com/ipitchford/weniger-delta-nonreal-poles (tag `v1.0.0-candidate`). It contains `code/verify_all.py` (Python 3; exact parts use only the standard library, the illustrative `numerical_*` sections use mpmath). Run

```
python3 -B -O code/verify_all.py results
```

It writes `results/verification.json` with sorted keys, no timestamps or paths, and floats rounded to eight significant figures, and stops with an error if any exact check fails. It checks: the corner identities and the discriminant identity (12) over $\mathbb{Q}[b, x, y]$; identity (3) coefficientwise over $\mathbb{Q}[b]$ for $n \le 3$, $k \le 8$; (8) over $\mathbb{Q}[b]$ for $3 \le k \le 12$; the rational form (4) against the definition (2) at 81 rational points; the explicit cubic and quartic examples (discriminants, Laguerre value, gcds, resultants, accuracy); the Legendre integrals; the cubic sharpness family; the all-shift identities of Proposition 5.1, the four real zeros of $Q_4^{(2,1)}$ for λ, and the identities for g_b in Corollary 5.2; the factorisation $(k-4)(4k-5)$ in (9); the persistence certificate of Theorem 6.3 including the Jensen polynomials $g_1, \ldots, g_{12}$; Table 3; and the natural-family tests and the Laguerre identity, including its constant, and the large-shift Jacobi example of §7. Finite checks supplement the proofs; they do not replace the all-parameter arguments.

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